

Baseline Hard Drug Use and Two-Year HAART Response in HIV-Infected Men

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1 Introduction

This study is a secondary analysis of data from the Multicenter AIDS Cohort Study (MACS), a prospective cohort investigating HIV-1 disease progression among homosexual and bisexual men in four U.S. cities. Among 715 HIV-infected men who initiated highly active antiretroviral therapy (HAART), we examined whether changes in clinical and health-related outcomes over the two years following treatment initiation differed according to baseline hard drug use.

Our primary hypothesis was that baseline hard drug use would be associated with attenuated treatment-related improvements. Specifically, in regression models with no hard drug use as the reference group, we expected hard drug use to be associated with a smaller decline in viral load and smaller increases in CD4+ T-cell count and physical and mental aggregate scores over 2 years of follow-up.

2 Methods

2.1 Exploratory Analysis

We first filtered the dataset to retain only baseline and year-2 observations of interest. Continuous variables were screened for extreme values. For example, CD4+ T-cell count was required to be positive. BMI values below 0 or greater than 100 were set to missing.

The final analytic sample size was determined by complete-case analysis, excluding all individuals with missing values in any of the four outcomes, the exposure variable, or the six covariates.

Categorical variables were recoded according to the principal investigator's guidance. BMI was categorized using 18.5 and 24.9 as cutoffs into underweight, normal, and overweight. Education was dichotomized into college or above versus incomplete college or below. Race was categorized as Non-Hispanic White versus other. Smoking status was categorized as current smoker versus non-current smoker.

The outcomes were defined as change from baseline to two years in viral load, CD4+ T-cell count, aggregate physical and mental quality-of-life score. Prior to computing change scores, histograms of baseline values were examined. If a variable was highly right-skewed, a $\log_{10}$ transformation was applied before computing the difference.

Pearson correlation matrices were examined to assess potential multicollinearity among outcomes and covariates.

We then stratified participants by baseline hard drug use and generated a summary table. For continuous variables, means and standard deviations were calculated, and two-sample t-tests were performed. For categorical variables, counts and proportions were reported, and chi-squared tests were conducted.

2.2 Formal Analysis

We modeled the four change scores as continuous outcomes using linear regression, adjusting for age, race, education, smoking status, and BMI. Adherence was additionally included in secondary models to assess consistency of the main effect.

Bayesian regression models were estimated using non-informative priors. Initial values were generated uniformly between -2 and 2. Four chains were run, each with 2000 warm-up iterations and 10000 sampling iterations.

Model diagnostics included evaluation of effective sample size (ESS), which was required to exceed 1000, assessment of convergence using the $\hat{R}$ statistic with a threshold below 1.01, and visual inspection of trace plots to ensure adequate convergence.

For each outcome, a frequentist linear model was also fitted using `lm()` in R to evaluate consistency between Bayesian and frequentist estimates.

For all eight Bayesian models, we summarized posterior means, 95% posterior intervals, posterior probabilities based on predefined clinical thresholds (0.5 for change in $\log_{10}$ viral load, -50 for change in CD4+ T-cell count, and -2 for both aggregate quality-of-life scores), and the widely applicable information criterion (WAIC).

Model comparison was conducted using WAIC together with assessment of consistency between Bayesian and frequentist estimates. Based on these evaluations and considerations of model parsimony, we selected the Bayesian models without adherence as the primary models.

Posterior density plots were generated for the primary models to visualize the distributions of the main effects.

2.3 Software

All analyses were conducted using R version 4.5.2. Bayesian models were implemented in Stan via the `cmdstanr` interface. Frequentist linear models were fitted using `lm()`. T-tests

and chi-squared tests were performed using `gtsummary`.

3 Results

3.1 Exploratory Analysis

After removal of extreme values and missing observations, the final analytic sample consisted of 463 individuals.

Viral load was highly right-skewed and therefore $\log_{10}$ transformed. After transformation, all four outcomes were approximately normally distributed.

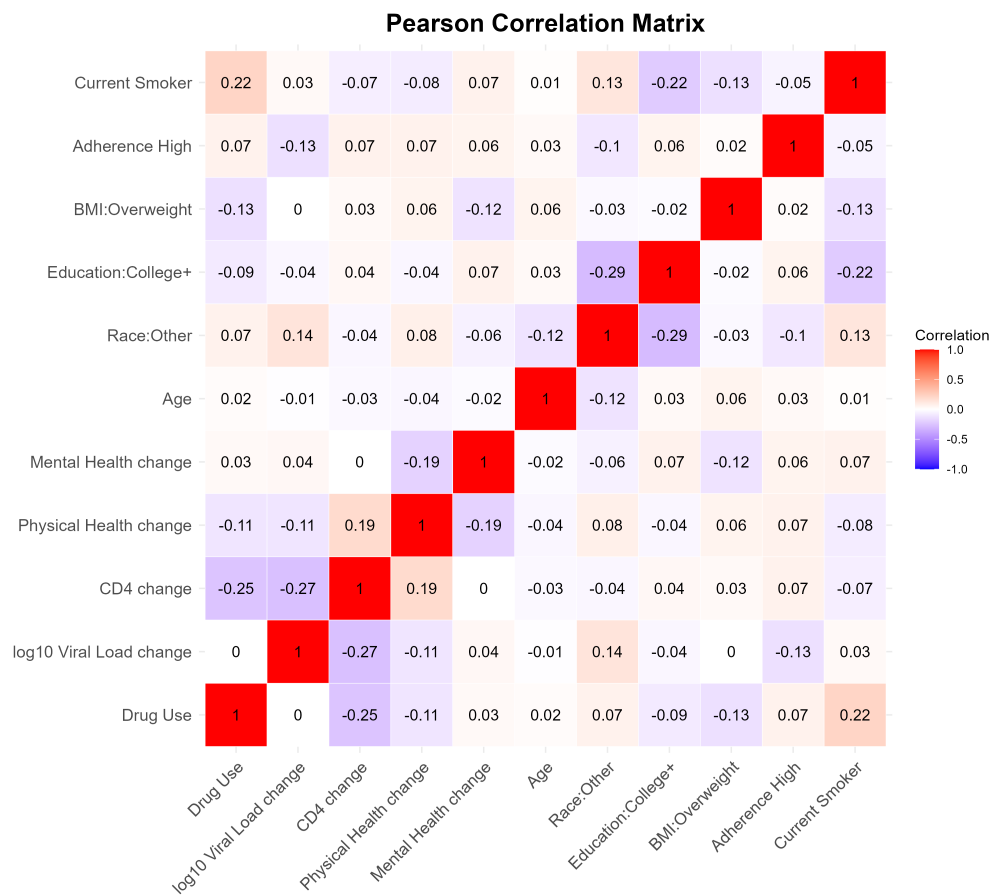


Figure 1: Correlation Matrix

Pearson correlation were all below 0.5, indicating no serious multicollinearity.

Table 1: Baseline Characteristics by Hard Drug Use

Characteristic	No (N = 4271)	Yes (N = 361)	p-value
Age (years)	43.10 (8.68)	43.89 (9.52)	0.6
Race/Ethnicity			0.2
Non-Hispanic White	277 (65%)	19 (53%)	
Other	150 (35%)	17 (47%)	
Education			0.073
College or higher	191 (45%)	10 (28%)	
No college degree or lower	236 (55%)	26 (72%)	
BMI category			0.019
Normal	205 (48%)	26 (72%)	
Overweight	208 (49%)	9 (25%)	
Underweight	14 (3.3%)	1 (2.8%)	
Smoking			<0.001
Current smoker	149 (35%)	27 (75%)	
Non-current	278 (65%)	9 (25%)	
Adherence to Treatment at Year 2			0.4
<75%	12 (2.8%)	0 (0%)	
100%	177 (41%)	18 (50%)	
75–94%	33 (7.7%)	1 (2.8%)	
95–99%	205 (48%)	17 (47%)	
Change in log ₁₀ Viral Load	-2.74 (1.22)	-2.72 (1.32)	>0.9
Change in CD4 count	181.64 (175.10)	11.18 (203.64)	<0.001
Change in Physical Health	-1.34 (8.17)	-4.80 (8.38)	0.022
Change in Mental Health	2.42 (11.75)	3.89 (15.65)	0.6

Baseline hard drug users had a higher proportion of normal BMI (72% vs 48%, $P = 0.019$) and were more likely to be current smokers (75% vs 35%, $P < 0.001$).

Among the four outcomes, only change in CD4+ T-cell count and change in physical score were statistically significant in unadjusted comparisons. Hard drug users experienced a smaller increase in CD4+ count (11 vs 182, $P < 0.001$) and a larger decline in physical score (-5 vs -1, $P = 0.022$).

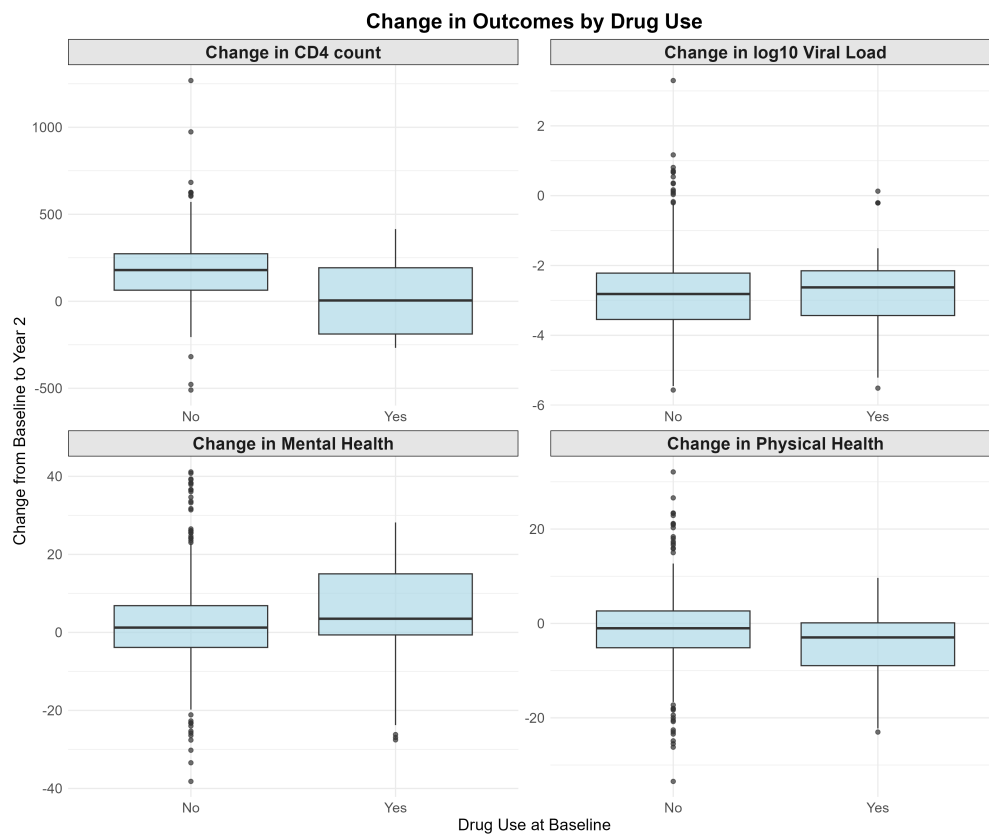


Figure 2: Boxplots of Outcomes by Baseline Hard Drug Use

The boxplots demonstrated similar trends, with noticeable differences in the distribution and median positions of CD4+ T-cell count and physical health score between the two groups.

3.2 Formal Analysis

Table 2: Bayesian Model Results

Outcome	Posterior Estimate	95% Posterior Interval	Posterior P	WAIC
Δ Viral Load	-0.057	(-0.486, 0.375)	0.994	1513.3
Δ Viral Load w/ adh	-0.005	(-0.437, 0.427)	0.989	1511.6
Δ CD4 Cell Count	-152.8	(-211.6, -92.8)	<0.001	6125.9
Δ CD4 Cell Count w/ adh	-157.0	(-216.4, -97.1)	<0.001	6120.3
Δ Physical Score	-3.12	(-5.99, -0.24)	0.222	3269.8
Δ Physical Score w/ adh	-3.31	(-6.16, -0.40)	0.182	3264.1
Δ Mental Score	0.48	(-3.74, 4.72)	0.875	3627.9
Δ Mental Score w/ adh	0.18	(-4.13, 4.48)	0.840	3631.1

All eight models demonstrated excellent convergence, with bulk and tail ESS exceeding 20000 and $\hat{R}$ well below 1.01, and visual inspection of the trace plots revealed no anomalies.

Baseline hard drug use was not significantly associated with change in $\log_{10}$ viral load or mental score.

For CD4+ T-cell count, baseline hard drug users experienced on average 150–160 fewer cells of increase compared to non-users. The 95% posterior interval ranged approximately from -210 to -90, not covering 0 or the clinically meaningful threshold of -50. The posterior probability that the decrease exceeded the clinically meaningful threshold was essentially 1, indicating a statistically and clinically significant effect.

For physical score, hard drug users experienced a smaller improvement (approximately -3 points), with a 95% interval from roughly -6 to -0.2. The posterior probability that the decrease exceeded the clinically meaningful threshold was approximately 0.8,

indicating that this effect is clinically significant, or roughly significant.

Results were highly consistent between models with and without adherence, as well as between Bayesian and frequentist approaches.

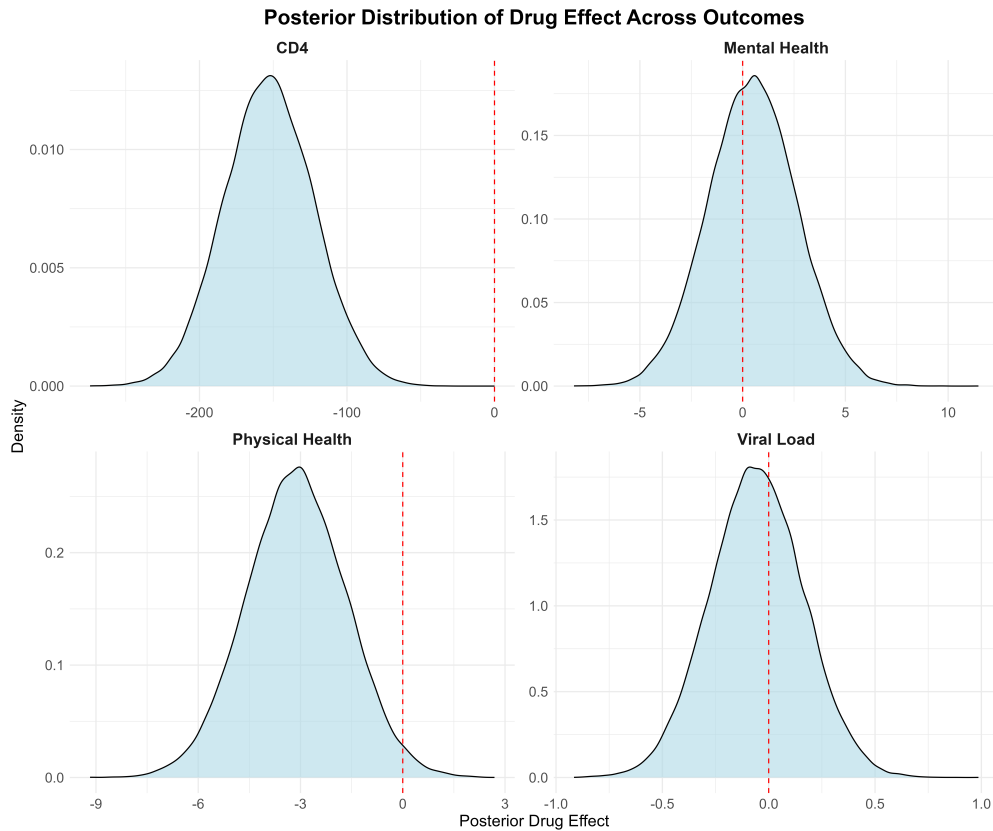


Figure 3: Posterior Density Plots of Main Effects

From the density plots, the posterior distributions of the main effects in all four models appear approximately normal and unimodal, without any evidence of multiple modes, indicating that the estimates are well-behaved and the models have likely converged reliably.

4 Discussion

Baseline hard drug use was associated with significantly attenuated recovery of CD4+ T-cell count and smaller improvements (or greater declines) in physical quality-of-life score

over two years following HAART initiation, compared to non-users. Notably, physical scores declined over two years across all groups, which was unexpected.

Several limitations should be acknowledged. Missing data were handled via complete-case analysis, potentially introducing bias if data were missing not at random. Only 7.7% of participants reported baseline hard drug use, limiting statistical power. Variable selection was limited, and additional confounders may warrant further exploration.

Appendix: Code and Data

All code used to generate the analyses and figures in this report is available on GitHub. The specific R Markdown file for this project can be accessed at: https://github.com/Guangyu-Qiu/Adv_method/blob/master/Project_1/Code/Project1.Rmd. The Stan files can be found at https://github.com/Guangyu-Qiu/Adv_method/tree/master/Project_1/Code/stan.

The full repository containing all project files is available at: https://github.com/Guangyu-Qiu/Adv_method.git.

No additional generated data or report files were produced for this project; all analyses are fully reproducible using the files in the repository.

The data are stored locally on my computer at `C:/Users/guang/OneDrive/Documents/Adv_method/Project_1/Data`, but I manage them using the `here` package and R Project, which makes access much simpler.